

Homework IoT - Exercise 1

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Consider the following pseudocode for an ESP32-based vibration monitoring node:

```
1 // Global timer handle
2 declare timer_handle as esp_timer_handle_t
3
4 function setup_accelerometer():
5     initialize_accelerometer(sampling_rate = 100 Hz)
6
7 function setup_timer():
8     create_periodic_timer(callback = process_window,
9                             period = 10 seconds)
10
11 function process_window(arg):
12     vibration_peaks = count_vibration_peaks(last_10_seconds)
13
14     if vibration_peaks == 0:
15         payload = create_message(size = 512 bytes)
16     else if vibration_peaks == 1:
17         payload = create_message(size = 2 KB)
18     else:
19         payload = create_message(size = 4 KB)
20
21     send_payload(payload)
22
23 function app_main():
24     setup_accelerometer()
25     setup_timer()
26
27     loop forever:
28         delay(100 ms)
```

Assume that the number of vibration peaks detected in each 10-second window follows a Poisson distribution with an average rate:

$$\lambda = 0.4 \text{ peaks/window} \quad (1)$$

The system is operated using IEEE 802.15.4 in beacon-enabled mode, utilizing the CFP only. Assume:

- $R = 250 \text{ kbps}$
- $L = 128 \text{ bytes}$

Scenario Constraints

Assume that:

- one packet fits exactly into a single CFP slot.
- the CFP is dimensioned according to the expected traffic load.

- the reporting period is set to 10 seconds.
- the monitoring system is composed of 1 PAN coordinator and 3 vibration monitoring nodes

1 Question 1

Compute the Probability Mass Function of the output rate: $P(r=r_0), P(r=r_1), P(r=r_2)$ where r_0, r_1 , and r_2 are the output rates when the node detects 0, 1, or more than 1 vibration peaks, respectively.

Since the number of vibration peaks in a 10-second window follows a Poisson distribution with $\lambda = 0.4$ we can use this formula

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad (2)$$

to calculate the probabilities for each peak.

For 0 peaks ($k = 0$):

$$P(r = r_0) = \frac{e^{-0.4} \cdot 0.4^0}{0!} \approx 0.6703$$

For 1 peak ($k = 1$):

$$P(r = r_1) = \frac{e^{-0.4} \cdot 0.4^1}{1!} \approx 0.2681$$

For more than 1 peak ($k > 1$): since the sum of all probabilities must equal 1, we can subtract the probabilities of 0 and 1 from 1:

$$\begin{aligned} P(r = r_2) &= 1 - (P(X = 0) + P(X = 1)) \\ &= 1 - (0.6703 + 0.2681) \approx 0.0616 \end{aligned}$$

2 Question 2

Based on the output rate PMF, compute a consistent CFP slot assignment for the system.

To define a consistent CFP slot assignment, we first need to compute the Beacon Interval (BI) using the standard formula $BI = \frac{L}{r_{min}}$.

The minimum output rate r_0 (0 peaks) is:

$$r_0 = \frac{512 \text{ bytes}}{10 \text{ seconds}} = 409.6 \text{ bps}$$

After computing the minimum output rate we can calculate the Beacon Interval:

$$BI = \frac{128 \text{ B}}{51.2 \text{ B/s}} = 2.5 \text{ seconds}$$

Since the reporting period is 10 seconds, the payload generated by a node is distributed over 4 Beacon Intervals ($10s/2.5s = 4$).

To dimension the slots per single BI we divide the total payload by 4, and then divide by the packet size L which is equal to 128 bytes:

- 0 Peaks (512 bytes in 10s): $128 \text{ bytes/B I} \rightarrow \frac{128}{128} = 1 \text{ slot}$
- 1 Peak (2000 bytes in 10s): $500 \text{ bytes/B I} \rightarrow \frac{500}{128} = 3.906 \rightarrow \lceil 3.906 \rceil = 4 \text{ slots}$
- >1 Peaks (4000 bytes in 10s): $1000 \text{ bytes/B I} \rightarrow \frac{1000}{128} = 7.8125 \rightarrow \lceil 7.8125 \rceil = 8 \text{ slots}$

To guarantee transmission in the worst-case scenario (>1 peaks), we must assign 8 slots per node per BI. Since the monitoring system is composed of 3 vibration monitoring nodes, the total CFP will require $8 * 3 = 24$ slots.

3 Question 3

Compute:

- T_s
- N_{slots} in the CFP
- T_{active}
- T_{inactive}
- The duty cycle of the system

3.1 T_s

T_s is found by dividing the number of bits in a packet by the nominal bit rate R .

$$T_s = \frac{L}{R} = \frac{(128 \times 8) \text{ bits}}{250,000 \text{ bps}} = 0.004096 \text{ seconds}$$

$$T_s = 4.096 \text{ ms}$$

3.2 N_{slots} in the CFP

This is calculated in point 2 and it's equal to 24 slots.

3.3 T_{active}

The active part of the superframe includes all the slots in the CFP plus one additional slot allocated for the beacon, which brings the total number of active slots to 25 slots per BI. To get the duration of the active part we multiply that with the duration of a single slot (T_s):

$$T_{\text{active}} = 25 \text{ slots} \cdot 4.096 \text{ ms/slot}$$

$$T_{\text{active}} = 102.4 \text{ ms}$$

3.4 T_{inactive}

The duration of the inactive part is the Beacon Interval minus the active duration (T_{active}).

$$T_{\text{inactive}} = BI - T_{\text{active}}$$

$$T_{\text{inactive}} = 2500 \text{ ms} - 102.4 \text{ ms}$$

$$T_{\text{inactive}} = 2397.6 \text{ ms}$$

3.5 Duty cycle of the system

The duty cycle is the ratio of the active duration to the total Beacon Interval.

$$\eta = \frac{T_{\text{active}}}{BI}$$

$$\eta = \frac{102.4 \text{ ms}}{2500 \text{ ms}} = 0.04096$$

Duty Cycle (η) $\approx 4.096\%$

4 Question 4

How many additional vibration monitoring nodes can be added while keeping the duty cycle below 10%?

By multiplying the maximum possible duty cycle value by the total interval (BI), we can find the maximum active time tolerated by the system:

$$T_{active,max} = 0.10 \cdot BI = 0.10 \cdot 2500 \text{ ms} = 250 \text{ ms}$$

After calculating that, we must convert it into the maximum number of equivalent slots. Knowing that the duration of a single slot (T_s) is 4.096 ms, we divide the maximum active time by it and round down, since we cannot transmit in a fraction slot:

$$N_{active,max} = \lfloor \frac{T_{active,max}}{T_s} \rfloor = \lfloor \frac{250 \text{ ms}}{4.096 \text{ ms/slot}} \rfloor = 61 \text{ slots}$$

The result means that the system can have a maximum of 61 active slots. Subtracting the 1 slot mandatory for the beacon, we have 60 slots available for the CFP. As we calculated it before, we know that each new node will require in the worst case scenario 8 slots to guarantee transmission. This means that to calculate the number of additional nodes we have to divide the slots obtained by the slots needed by a single node, which is:

$$\text{Additional nodes} = \lfloor \frac{60 \text{ slots}}{8 \text{ slots/node}} \rfloor = \lfloor 7.5 \rfloor = 7 \text{ nodes}$$

Since there are already 3 nodes in the network, the number of additional nodes we can add is 4.